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EXERCISE 01 - SQL FOUNDATIONS

Q1 Retrieve all columns from the employees table.

1. SELECT *
FROM employees;

id	first_name	last_name	department	Salary	hire_date	City
1	Alice	Green	IT	70 000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	CapeTown
3	cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

Q2 Find all unique departments.

2 SELECT DISTINCT department
FROM employees;

department
IT
HR
Finance
Marketing

Q3 Retrieve first and last names ordered by salary descending.

3. SELECT first_name, last_name
FROM employees
ORDER BY Salary DESC;

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first_name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

Q4 Retreve the top 3 highest-paid employees.

```
4 SELECT id, first_name, last_name, salary
  FROM employees
  ORDER BY salary DESC
  LIMIT 3;
```

id	first_name	last_name	Salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

Q5 Find employees in the IT department.

```
5. SELECT id, first_name, last_name, department
   FROM employees
   WHERE department = 'IT';
```

id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

Q6 Find employees in Finance with Salary > 60 000.

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6. SELECT id, first-name, last-name, department, salary
FROM employees
WHERE department = 'Finance'
AND Salary > 60000;

id	first-name	last-name	department	Salary
3	Cathy	Zulu	Finance	65000

Q7 Find employees in HR or Marketing.

7. SELECT id, first-name, last-name, department
FROM employees
WHERE department = 'HR'
OR department = 'Marketing';

id	first-name	last-name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

Q8 Find employees not in IT

8. SELECT id, first-name, last-name, department
FROM employees
WHERE department NOT = 'IT';

id	first-name	last-name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

Q9 Find employees in IT, HR, or Finance using IN

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9. SELECT id, first_name, last_name, department
FROM employees
WHERE department IN ('IT', HR', 'Finance');

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

Q10 Find employees in IT with salary > 65000 and city Johannesburg

10. SELECT id, first_name, last_name, department, salary, city
FROM employees
WHERE department = 'IT'
AND salary > 65000
AND city = 'Johannesburg';

id	first_name	last_name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg